CHK2 kinase: cancer susceptibility and cancer therapy – two sides of the same coin?

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Abstract | In the past decade, CHK2 has emerged as an important multifunctional player in the DNA-damage response signalling pathway. Parallel studies of the human *CHEK2* gene have also highlighted its role as a candidate multiorgan tumour susceptibility gene rather than a highly penetrant predisposition gene for Li–Fraumeni syndrome. As discussed here, our current understanding of CHK2 function in tumour cells, in both a biological and genetic context, suggests that targeted modulation of the active kinase or exploitation of its loss in tumours could prove to be effective anti-cancer strategies.

Cell-cycle checkpoint
A molecular check in the cell cycle to prevent initiation of the next phase in case the DNA is damaged or another condition endangers accurate and safe cell division.

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Human cells activate the DNA-damage response during early cancer lesions, and this inducible mechanism is thought to prevent or delay genetic instability and tumorigenesis, thus acting as a barrier against cancer development[1,2]. To maintain genome integrity, the cell relies on complex signalling networks to coordinate cell-cycle checkpoints that, in response to DNA damage, allow for the cell cycle to arrest and DNA repair to proceed, or activate senescence or cell death. It is thought that individuals with defects in their DNA-damage response signalling pathways lose their natural protection against tumorigenesis and are more susceptible to cell transformation and cancer.

The checkpoint kinase CHK2 is central to transducing the DNA damage signal. Although mutations in *CHEK2* (the gene encoding CHK2) do not account for the cancer-predisposing Li–Fraumeni Syndrome (LFS) as originally thought, rare germline mutations have been detected with high incidence in a number of familial cancers and rare somatic mutations have been reported in some tumours[3–8]. It has therefore been proposed that *CHEK2* is a multiorgan cancer susceptibility gene that functions in the barrier to tumorigenesis to maintain genomic stability.

Although genotoxic treatments that cause DNA damage, such as chemotherapy or radiotherapy, have been used in the clinic for several decades with the aim of killing cancer cells, a large number of tumours fail to respond to these treatments. The functional availability of the molecular network that responds to DNA damage is likely to direct how a patient's tumour responds to therapeutic DNA damage. The initial discovery of CHK2 (BOX 1) generated much hope and anticipation regarding its potential as a therapeutic target in the treatment of cancer. It has been proposed that checkpoint inhibitors, such as drugs that inhibit CHK2 function, combined with genotoxic agents could have therapeutic value in tumours that already possess other defects in the DNA-damage response, such as p53 deficiency, by preventing cell-cycle arrest and DNA repair and by activating cell death[9].

So, why would you wish to therapeutically target a kinase that acts as a barrier against tumorigenesis? Using the analogy of two sides of the same coin, there are two types of tumours: those that express wild-type and functional CHK2 and those in which its function is diminished or eliminated. For wild-type CHK2 tumours that possess other defects in checkpoint and repair pathways, short-term pharmacological intervention at the level of CHK2 could have therapeutic value, as inhibition of CHK2 in these tumours might be a lethal event. Alternatively, in those tumours where CHK2 function is diminished or eliminated, inhibition of other proteins involved in checkpointand repair pathways might be lethal[10].

Here we review the biology and genetics of CHK2, discuss how this kinase may well be therapeutically exploited for the treatment of cancer and examine the current status of CHK2 drug discovery.

Cellular regulations and functions of CHK2

Since the discovery of CHK2, much effort has focused on unravelling the regulation of its activity, as this is crucial to understanding how tumour cells respond to DNA damage. Depending on the nature of the damage,

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At a glance

- CHK2 is a versatile and multifunctional kinase that regulates the cell's response to DNA damage by phosphorylating a number of distinct cellular substrates.
- CHK2 can prevent tumour progression by averting genomic instability through DNA repair and, if this is not possible, by causing the cell to senesce or die.
- Human genetic studies clearly show that *CHEK2* is a multiorgan tumour susceptibility gene, but current evidence indicates that *CHEK2* on its own does not predispose to cancer.
- A potential therapeutic approach in patients whose tumours harbour *CHEK2* mutations may be treatment with inhibitors of other proteins that are involved in DNA-repair pathways, inactivation of which may be lethal in combination with a loss of *CHEK2*.
- Looking at the other side of the coin, in cancer patients with a functional CHK2 protein a key issue is defining how this kinase manages to elicit distinct cellular outcomes such as cell survival through DNA repair versus apoptosis or senescence.

both CHK2 and the functionally related CHK1 kinase can be activated by the ataxia telangiectasia mutated (ATM) and ATM- and Rad3-related (ATR) kinases. These two kinases act as transducers of the DNA-damage signal as they are downstream of specific complexes of protein sensors that detect the insult to DNA[9,11,12] (FIG. 1). Both structural and biochemical studies of CHK2 have revealed the complexity of its activation, which is initiated by ATM phosphorylation of CHK2 on T68 (FIGS 1,2). Both CHK2 and CHK1 act as amplifiers of the DNA-damage signal, phosphorylating a multitude of substrates involved in the DNA-damage response. As can be seen in FIG. 1, CHK2 and CHK1 share a number of overlapping substrates, leading to the initial proposal that they could be functionally redundant. It is now clear, however, that they have distinct roles in directing the cell's response to DNA damage (BOX 2). The response of CHK2 is driven through phosphorylation of a number of substrates, described below, whose functions encompass the cellular outcomes subsequent to CHK2 activation.

DNA repair. Genome integrity relies on the cell accurately repairing damaged DNA, and mutations must not be transmitted to daughter cells as this may favour tumour progression. There is increasing evidence that CHK2 regulates DNA repair. Following CHK2 phosphorylation on S988, BRCA1 is released from nuclear foci and functions within the error-free homologous recombination repair pathway, while repressing the error prone non-homologous end joining (NHEJ) repair pathway[13–16]. In addition, CHK2 phosphorylates the transcription factor FOXM1, which enhances its stability, thereby promoting the expression of BRCA2 and base excision repair factor X-ray cross-complementing group 1 (XRCC1), which operate within the homologous recombination repair and the base-excision repair pathways, respectively[17]. It is therefore possible that when DNA-damaging agents are combined with CHK2 inhibitors, DNA repair will be impaired and tumour cells that are unable to arrest accumulate irreparable DNA damage and consequently undergo cell death.

Cell-cycle arrest. In order to allow for repair to proceed, cells delay DNA synthesis and cell division following DNA damage. The original studies that identified human CHK2 demonstrated that it can phosphorylate the CDC25C phosphatase, which is required for the activation of cyclin-dependent kinase (CDK) complexes that regulate cell-cycle progression. Phosphorylation of CDC25C on the inhibitory residue S216 promotes binding of the 14-3-3 protein and nuclear export, thus causing a G2/M delay and preventing cells from entering mitosis[18–20]. In addition, CHK2 also contributes to phosphorylation of CDC25A, another CDK phosphatase, on S123, S178 and S292, promoting its proteasomal degradation and causing G1 and S-phase delay after exposure to ionizing radiation (IR)[21,22]. Surprisingly, analysis of CHK2-deficient mice has revealed that the initiation and maintenance of G2 arrest, initiation of G1 arrest and S-phase blockage following IR exposure all occur normally in the absence of CHK2, suggesting that although CHK2 has a role in these functions, it is not essential[23]. One explanation for this is that CHK1, which is also a CDC25A and CDC25C kinase, might compensate for the loss of CHK2 and activate these checkpoints.

CHK2 has been implicated in mediating both G1/S and G2/M cell-cycle arrest in a distinct pathway through p53. The p53 transcription factor responds to numerous cellular stresses and eliminates cells carrying oncogenic lesions or damaged DNA, thus preventing tumour development. The high frequency of p53 mutations in human cancers highlights its role as a tumour suppressor. Whereas ATM phosphorylates S15 on p53, both CHK1 and CHK2 phosphorylate p53 on S20, which disrupts its association with the ubiquitin ligase MDM2, thus promoting its stability[24–27]. It has also been suggested that p53 expression might be increased by the apoptosis-antagonizing transcription factor (AATF, also known as CHE-1), which is phosphorylated and activated by CHK2 and ATM in response to DNA-damaging agents[28]. Direct CHK2-mediated phosphorylation of p53 also promotes its association with the histone deacetylase p300 and positively regulates its transcriptional activity[29]. Furthermore, a number of other p53 residues that are phosphorylated in response to DNA damage, in particular S366 and T387, have been shown to be phosphorylated by CHK2 and CHK1, serving to regulate the levels of acetylation and hence the activity of p53 (REF. 30). Recently, CHK2 has been shown to be essential for the phosphorylation of another negative regulator of p53, MDM4 (also known as MDMX). Phosphorylation of MDM4 on S367 promotes its binding to 14-3-3 and degradation by MDM2, thereby increasing p53 stability and activity in response to DNA damage[31,32].

Although biochemical studies have demonstrated that CHK2 regulates p53, it is presently unclear whether CHK2 is required to activate p53-mediated cell-cycle arrest in response to IR. Indeed, numerous studies using mouse knockout mutants and CHK2-deficient human cell lines have reported differing results (BOX 3). This may be due to the variations of techniques used or to intrinsic differences between species or cell lines used for these studies.

Li–Fraumeni Syndrome (LFS). A rare syndrome characterized by a familial cluster of very early onset cancers at multiple sites, including sarcoma, breast, brain and adrenocortical tumours. Definition of a classical LFS family: a sarcoma at <45 years, one first-degree relative with cancer at <45 years and one first- or second-degree relative in the same lineage with cancer at <45 years.

Non-homologous end joining repair (NHEJ). Unlike homologous recombination repair, NHEJ rejoins broken ends of DNA following double-strand breaks without using a homologous DNA template and can therefore be accompanied by loss of nucleotides and errors.

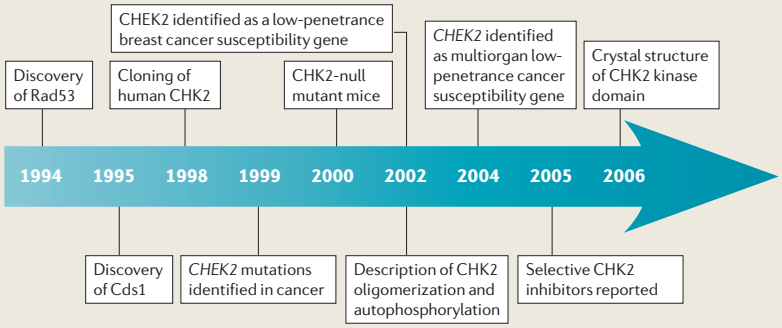
Base-excision repair (BER). BER replaces non-bulky damage of single bases that have been altered by alkylation, oxidation or deamination.

14-3-3 A group of proteins that bind to phospho-proteins and regulate their action mainly by controlling their subcellular localization.

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Box 1 | **Brief history of CHK2 research**

Like many key cell-cycle regulators, CHK2 was first identified in the budding yeast *Saccharomyces cerevisiae* as Rad53 and soon after in the fission yeast *Schizosaccharomyces pombe* as Cds1 (REFS 120,121) (see Timeline). These early studies revealed that Rad53 and Cds1 are serine/threonine kinases that monitor DNA replication and activate cell-cycle arrest in response to DNA damage. Elledge and co-workers[122] first identified the human CHK2 homologue, the activation of which was shown to require ataxia telangiectasia mutated (ATM) function following ionizing radiation but not after ultraviolet- or hydroxyurea-induced replication block[122–126]. In 2002, *CHEK2* 1100delC was found to be a low-penetrance breast-cancer susceptibility allele in individuals that do not carry mutations in *BRCA1* or *BRCA2* (REF. 61). An association of mutations in *CHEK2* with prostate cancer was shown in 2003 (REF. 83). *CHEK2* was reported as a multiorgan cancer susceptibility gene in 2004 (REF. 68). In addition, the first mouse mutant for CHK2 was obtained in 2000 (REF. 127). The first highly selective ATP-competitive CHK2 inhibitors were reported in 2005 (REF. 118). This was soon followed by the publication of the X-ray crystal structure of the CHK2 kinase domain, which will aid in the development of additional chemical classes of CHK2-selective inhibitors[114].



It is therefore possible that although CHK2 is implicated in activating p53-mediated cell-cycle arrest, this function may not be essential; the exact role of CHK2 in blocking cell-cycle progression remains to be clarified.

Apoptosis. When the amount of damage to a cell is not repairable it can trigger apoptosis, and there is now compelling evidence that CHK2 participates in the activation of this process. Indeed, genetic deletion of the mouse *Chek2* gene obliterates p53-dependent cell death following radiation[23,33–35]. Moreover, the DNA-dependent protein kinase, DNA-PK (also known as PRKDC), which functions in the NHEJ pathway, can act synergistically with CHK2 to activate p53-dependent apoptosis in response to DNA damage, as well as activating CHK2 itself independently of ATM[36,37]. Importantly, these studies indicate that, because of potential genetic differences between normal tissues and tumour cells and because of the pro-apoptotic role of CHK2, its selective inhibition might have a radioprotective effect in normal tissues and might therefore help to improve the therapeutic index of radiotherapy. Such a principle remains to be demonstrated for chemoprotection. By contrast, blocking CHK2 function in p53-deficient human cells increases the level of apoptosis following radiation[38]. As most tumour cells are defective in one or more checkpoints, inhibition of the remaining CHK2-mediated checkpoint in combination with radiation might increase irreparable damage and tumour cell death.

CHK2 can also promote apoptosis by interacting with a number of other substrates. In particular, in response to etoposide-induced double-strand breaks, CHK2 phosphorylates S364 of the E2F1 transcription factor (from the E2F family), resulting in its stabilization, transcriptional activation and induction of apoptosis through a p53-independent mechanism[39]. Interestingly, expression of CHK2 is positively regulated by E2F1, which also activates ATM, thus increasing the apoptotic activity of p53 (REFS 40,41). CHK2 also promotes apoptosis independently of p53 in response to radiation by phosphorylating S117 of the tumour suppressor promyelocytic leukaemia (PML). PML mediates multiple pro-apoptotic pathways and its disruption promotes the development of acute promyelotic leukaemia. PML can also promote autophosphorylation and activation of CHK2 (REFS 42,43).

In addition, the Polo-like kinases PLK1 and PLK3 interact with CHK2 and are involved in the regulation of centrosome stability[44,45], which is necessary to prevent unequal chromosome segregation leading to genetic alterations during mitosis. CHK2 also negatively regulates mitotic catastrophe by activating G2/M arrest and preventing entry into mitosis. This suggests that inhibition of CHK2 may sensitize tumour cells that have an impaired DNA-damage signal response to chemotherapeutic agents[46]. This correlates with the recent observation that CHK2 activation is required for the release from mitochondria of survivin (also known as BIRC5), which is thought to inhibit apoptosis in cancer cells and might confer radiation resistance in human cancer cells[47]. This suggests that inhibition of CHK2 might increase apoptosis in those tumour cells that are resistant to radiotherapy. It will therefore be most valuable to further assess how CHK2 regulates cell death in the presence or absence of extrinsic DNA damage within a specific cellular context, in particular p53 status, as this may modify the functions of CHK2 within the apoptotic pathways.

Senescence. Cellular senescence is a form of permanent cell-cycle arrest, and evasion of senescence is a common theme in cancer[48]. Recent studies have implicated CHK2 as an inducer of senescence in a number of cellular contexts. CHK2 is activated by telomere erosion, which results from ongoing replication, and induces replicative senescence through p53 and expression of its transcriptional target, the CDK inhibitor p21, in non-damaged human cells. This process is thought to represent an innate defence against tumour progression[49]. In addition, forced overexpression of CHK2 was shown to activate both apoptosis and senescence in human cancer cell lines in a p21-dependent but p53-independent manner without ectopic DNA damage[50,51]. As most human tumours are defective in p53, further work will help define whether this constitutes an attractive mechanism to specifically inhibit tumour cell growth. Finally, CHK2 is required for oncogene-induced senescence to prevent hyperproliferation and subsequent tumorigenesis, and thus functions as an anti-cancer barrier[1,52].

Therapeutic index
Comparison of the amount of a therapeutic agent that has the desired effect with the amount that causes toxic or side effects.

E2F
Family of transcription factors that regulate the expression of genes implicated in cell-cycle progression and apoptosis, and whose activity is repressed by the retinoblastoma (pRb) tumour suppressor.

Mitotic catastrophe
Another form of cell death that occurs during or just after mitosis and is characterized by micronuclei or multi-nuclei.

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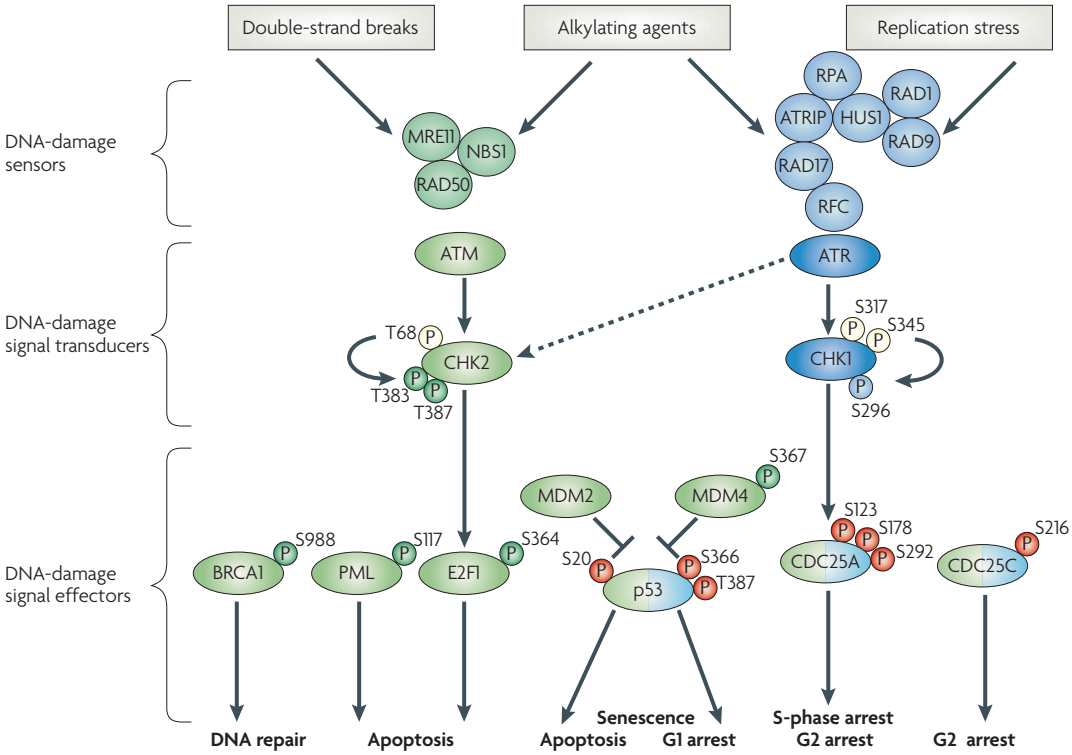


Figure 1 | **Schematic overview of the DNA-damage response signalling pathway.** Cells are constantly subjected to DNA damage and DNA breaks that arise either endogenously during normal cellular processes such as genome replication or exogenously by exposure to genotoxic agents, such as UV radiation, radiotherapeutics and chemotherapeutics. Damage to the DNA triggers the recruitment of specific damage sensor protein complexes. On the one hand, the MRN (MRE11–RAD50–NBS1) complex is required for the activation of ataxia telangiectasia mutated (ATM) in response to double-strand breaks (DSBs). On the other hand the ATM- and Rad3-related (ATR)-interacting protein (ATRIP) complex is recruited to sites of single-strand breaks and activates ATR. Specifically, the CHK2 pathway is activated by DSBs that occur either directly by exposure to ionizing radiation (and radiomimetic agents) or indirectly by topoisomerase II inhibitors. It can also be activated by replication-mediated DSBs as a result of base-pair excision generated by alkylating agents or single-strand breaks caused by topoisomerase I inhibition. Depending on the type of stress, activated CHK1 and CHK2 can phosphorylate a number of overlapping or distinct downstream effectors, which results in the activation of DNA repair, cell-cycle arrest, senescence or apoptosis. Symbols in green are CHK2-specific substrates, symbols in blue are CHK1-specific substrates and symbols in red are shared substrates. Downstream CHK1-specific substrates are not represented here.

Penetrance
The likelihood a given gene present in the germ line of an organism will result in disease.

Loss of heterozygosity
Classical tumour-suppressor genes are considered to be recessive. Thus, cells that contain one normal and one mutated form of a tumour-suppressor gene in the germ line are functionally normal. The condition that results in the loss of the remaining normal allele is known as loss of heterozygosity.

Dominant negative
A dominant-negative mutation occurs when the mutated gene product adversely affects the normal, wild-type gene product within the same cell.

Taken together, these data clearly show that a number of parallel, but not necessarily exclusive, pathways form the DNA-damage response, which is essential in cancer prevention, and that CHK2 is an important player in this response. Indeed, genetic studies in the germ line and in tumours indicate that CHK2 has a role in tumorigenesis.

The role of *CHEK2* in cancer

The cloning in 1998 of human *CHEK2* and the identification of its link to the DNA-damage response, led to a search for *CHEK2* mutations in both somatic and hereditary human cancers. Since then, there has been an avalanche of publications on this topic from which it is now clear that *CHEK2* is indeed a cancer susceptibility gene, but not a tumour suppressor gene in the classical sense.

CHEK2 and LFS. The first indication of the role of *CHEK2* in cancer came from a study that reported the presence of germline mutations in *CHEK2* in families with LFS[53]. LFS is a rare highly penetrant familial cancer syndrome[54,55]. About 75% of classical LFS families have a heterozygous germline mutation in the gene encoding p53, *TP53* (REFS 56,57). These individuals have an 85% risk of developing cancer over their lifetime and a 42% risk between birth and the age of 16 years[58]. Tumour initiation in some of these families is associated with a loss of heterozygosity (LOH), a phenomenon first described by Knudson for inherited classical tumour suppressor genes, which in this case results in a total loss of function of *TP53* (REF. 59). In other families, it has been hypothesized that the germline *TP53* mutation is either dominant negative or has a gain of function such that there is no selective pressure in these tumours for loss of the remaining wild-type *TP53* allele[57,60].

However, a subset of LFS families do not harbour germline *TP53* mutations. Given the potential role of CHK2 and CHK1 in the mammalian G2 checkpoint, a study was undertaken to determine whether mutations were present in these genes in LFS families[53].

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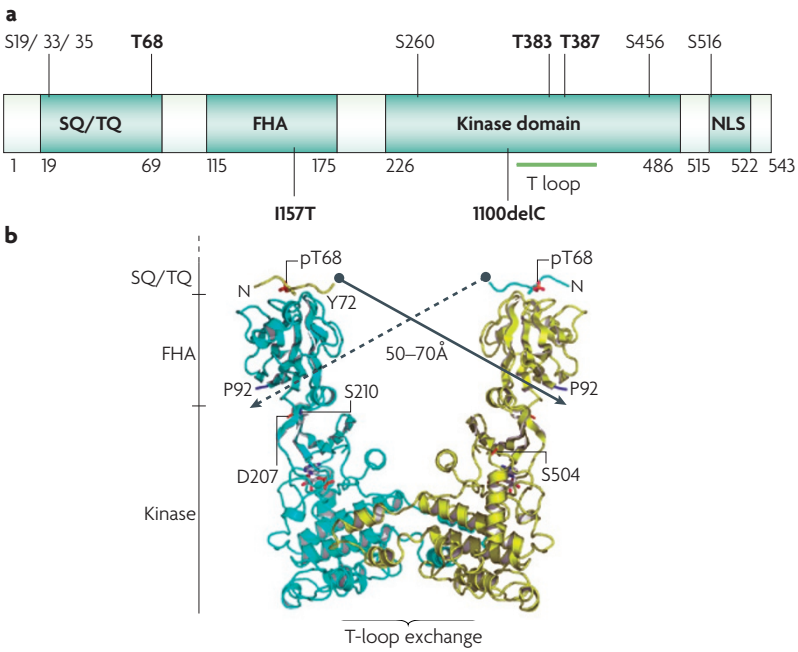


Figure 2 | **Structure and activation of human CHK2 protein. a** | The 543 amino-acid CHK2 protein is characterized by an amino-terminal domain rich in serine or threonine residues followed by glutamine (SQ/TQ motif – the consensus site for phosphoinositide-kinase-related kinases (PIKKs) such as ataxia telangiectasia mutated (ATM) and ATM- and Rad3-related (ATR)), a forkhead-associated (FHA) domain, which typically binds phosphothreonine residues, and a carboxy-terminal kinase domain that contains the activation T loop followed by a nuclear localisation signal (NLS). On the top of the figure the main residues that are phosphorylated to regulate the function of CHK2 are shown in bold, along with other residues that are phosphorylated in response to DNA damage. The main mutations in CHK2 in human tumours are shown on the bottom. Following treatment with ionizing radiation (IR), ATM phosphorylates CHK2 on T68 (REFS 126,132,133). Although the related ATR kinase can also phosphorylate CHK2 *in vitro*, little is known about its role in activating CHK2 in the cell[126,134]. Phosphorylation on T68 and subsequent full activation of CHK2 was recently shown to require priming phosphorylation on adjacent residues by Polo-like kinase 3 (PLK3) and the dual-specificity tyrosine and serine/threonine kinase TTK/hMPS1. Additionally TTK appears to phosphorylate T68 (REFS 135,136). Phosphorylation of T68 promotes the binding of the N-terminal SQ/TQ-rich cluster of one CHK2 molecule with the FHA domain of another CHK2 molecule. This dimerization is essential for full CHK2 activation by *trans*-autophosphorylation of T383 and T387 in the T loop of the kinase domain[137–140]. Both type 2A protein phosphatase (PP2A) and the PP2C phosphatase, PPM1D/WIP1, bind to and dephosphorylate T68 on CHK2, thus providing a recovery mechanism from DNA damage[141–143]. Although T68 phosphorylation is necessary for initial activation of CHK2, it is not required for its subsequent kinase activity either as a dimer or a monomer once the T loop is phosphorylated[138,144]. Moreover, a number of other residues on CHK2 are phosphorylated following IR, although their role in regulating CHK2 activity is presently unclear[140,145–147]. However, phosphorylation of S456 was recently shown to regulate the stability of CHK2 in response to DNA damage[148]. **b** | Schematic model of a full-length CHK2 dimer[114,150]. This model shows the atypical dimeric arrangement for CHK2, allowing exchange of the T loops and thus providing a mechanism by which dimerization-driven activation of CHK2 by *trans*-phosphorylation occurs[114]. Part **b** is modified, with permission, from REF. 114 © (2006) Nature Publishing Group.

The identification of three different *CHEK2* mutations, I157T, 1100delC and 1422delT, in one or more individuals from each of three separate LFS families led to the proposal that *CHEK2* is one of the alternative genes predisposing to LFS[53]. However, it was subsequently found that 1422delT is a polymorphism in a non-processed pseudogene[3] and that both I157T and 1100delC are founder mutations present at varying frequencies in normal healthy individuals in different populations[4,5] (TABLE 1). Therefore 1100delC and I157T cannot predispose to highly penetrant LFS. Furthermore, analysis of *CHEK2* in LFS families has revealed only a few rare mutations, and these have not been demonstrated to co-segregate with cancer in families with LFS[8].

CHEK2 as a multiorgan cancer susceptibility gene. Further studies on the prevalence of 1100delC and I157T founder mutations in cancer families and other cancer cases compared with normal healthy individuals indicate with statistical significance that these mutations are in fact moderately penetrant cancer susceptibility mutations, increasing the risk of developing breast and prostate cancer[61–69]. I157T has also been demonstrated to increase the risks of developing ovarian, colorectal, kidney, thyroid and bladder cancers and leukaemias[68,70–73]. However, other studies have indicated that 1100delC and I157T probably act in synergy with other genes or factors to cause cancer[61–63,74,75]. For example, 1100delC is more common in patients with breast cancer who have a first-degree relative with cancer or a family history of this disease, and the mean age of cancer incidence in 1100delC carriers is lower than that in non-carriers in case–control studies[62,65,74,75]. The carriers of 1100delC have also been found to have an increased risk for bilateral breast cancer[62,76–79]. Interestingly, the highest percentage of carriers for these mutations is breast cancer patients who develop contralateral breast cancer after having received radiation therapy for their first breast cancer[76]. Additionally, 1100delC carriers have also been shown to have an increased risk of breast cancer after exposure to non-mammographic X-rays[80]. These data indicate that environmental DNA-damaging factors such as IR increase the risk of cancer in carriers.

All these findings are consistent with the hypothesis that *CHEK2* is a multiorgan cancer susceptibility gene that acts in synergy with other genes or factors to cause cancer. To support this hypothesis, a recent report shows that the risk from *CHEK2* mutations in prostate and colon cancer may be restricted to individuals with a particular genotype of *CDKN1B* (the gene encoding the CDK inhibitor, p27 (REF. 81)). These genes, however, do not include *BRCA1* and *BRCA2* (REFS 61–63) possibly because the biological mechanisms underlying the increased risk of breast cancer in *CHEK2* mutation carriers are already subverted in *BRCA1* or *BRCA2* mutation carriers, consistent with proteins participating in the same pathway[61] (FIG. 1).

Subsequent studies have identified three other founder mutations: S428F, present in the Ashkenazi Jewish population, has been shown to confer an increased risk of breast cancer[82]; IVS2 + 1G>A has been reported to increase the risk of breast, prostate and thyroid cancer[6,66,68,83]; and 5395del has been shown to increase the risk of breast and prostate cancer[5,84] (TABLE 1). Full analysis of *CHEK2* in families with prostate cancer, prostate cancer cases unselected for a family history and families with breast cancer has

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Box 2 | **Key similarities and differences between CHK1 and CHK2**

Despite similar nomenclature and some overlapping functions, the roles of CHK1 and CHK2 are distinct, as shown in the table below. In order to further advance therapeutic opportunities and applications, it is essential to clearly define the distinct functions of those two kinases. This will be facilitated by the generation of novel small-molecule inhibitors that are selective for one kinase over the other one.

Function	CHK1	CHK2
Involved in the response to DNA damage	Yes	Yes
Phosphorylates CDC25 phosphatases and p53	Yes	Yes
Structure	Distinct from CHK2	Distinct from CHK1
Protein half-life	<2 hours	>6 hours
Activated by single-strand DNA breaks only	Yes	No
Phosphorylates BRCA1, PML1 and E2F1	No	Yes
Knockout mice	Embryonic lethal	Viable
Knockout mouse embryonic fibroblasts	Defects in G2 checkpoint	No detectable defects in the G2 checkpoint
Mutations identified in the germline	None	Rare founder mutations identified in different populations with varying frequencies

identified some rare, small deletions and nonsense mutations that are predicted to result in a truncated or null protein, owing to nonsense mediated RNA decay, and several missense mutations[83,85–88]. Some of these missense CHK2 mutants have been shown to be unstable or to affect the activation of the encoded protein, indicating that they are likely to be pathogenic[85,89–91].

One interesting question regarding cancer susceptibility is whether a total lack of CHK2 function is required for tumorigenesis, that is, is it necessary to inactivate both alleles of *CHEK2?* LOH studies in tumours in association with inherited *CHEK2* mutations have shown no consistent findings. Some tumours show loss of the wild-type allele and some show loss of the mutant allele and in many cases there is no alleleic imbalance[6,63,85,86,92–96]. It is known that for other genes a mutant protein may behave in a dominant-negative fashion to abolish the function of the wild-type allele. However, it has been shown that a number of mutations in *CHEK2,* including 1100delC, are unlikely to act this way. Therefore it has been suggested that such mutations in *CHEK2* may participate in tumorigenesis by haploinsufficiency[90].

Somatic mutations in CHEK2 in cancer. A few groups have analysed *CHEK2* in somatic tumours of different types and report rare, infrequent mutations in all studies[93,97–102]. Some of these mutations are present in a heterozygous state and some show loss of the wild-type allele. Loss of the arm of chromosome 22q.13 where *CHEK2* is located has been reported in a significant proportion of breast, colorectal, ovarian and brain tumours, but has not been found to be usually associated with mutations in the remaining allele[63,101–104]. A large number of tumour-specific splice variants, in addition to the normal-length mRNA, have been identified in stage III breast tumours, and it has been suggested that these splice variants might lack CHK2 function or be mislocalized in the cytoplasm and that this may be another mechanism by which CHK2 is inactivated in tumours[105]. A reduced expression and, in fewer cases, total lack of CHK2 protein, has also been reported. In most cases this is not associated with mutations or methylation of the promoter region, suggesting other epigenetic or post-transcriptional regulation factors might be at work[63,78,93–95,104–107].

Therefore, in tumours from both germline mutation carriers and from non-carriers, genetic and protein expression studies suggest that a reduced level rather than total lack of CHK2 function is necessary for tumorigenesis and that *CHEK2* is a multiorgan cancer susceptibility gene. Interestingly, it has recently been reported that carriers of the I157T *CHEK2* missense mutation show a decreased risk of tobacco-related (both lung and upper areo-digestive) cancers, suggesting a protective effect[108]. Thus, the same *CHEK2* allele may increase susceptibility for some cancers but decrease it for others. Taken together, these data indicate that by the Knudson definition *CHEK2* does not behave like a classical tumour suppressor gene.

Exploiting CHEK2 genetics in cancer treatment. A key issue now is whether our current knowledge about *CHEK2* and tumour susceptibility can be used to help treat patients with cancer. Testing most cancer patients for mutations in *CHEK2* is probably not yet appropriate, as the evidence indicates that *CHEK2* on its own does not predispose to cancer. However, if it is correct that *CHEK2* mutations are associated with an increased risk of contralateral breast cancer and with breast cancer developing from benign tumours after radiotherapy, knowing the *CHEK2* status of such patients might help in their management. Finally, is it possible to exploit the loss of CHK2 function as a therapeutic strategy? One possibility may be the use of poly(ADP-ribose) polymerase (PARP) inhibitors in

Pseudogene
A defective copy of all the sequence of a functional gene or of portions of it.

Founder mutation
A founder mutation is a mutation in the germline DNA of one or more individuals who are founders of a distinct population.

First-degree relative
A first-degree relative is one of the following: a parent, a sibling or a child.

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cancer patients who carry somatic but not germline mutations in *CHEK2,* as it has been reported that tumour cells that are functionally deficient for CHK2 are sensitive to PARP inhibition[109]. Indeed, PARP inhibitors are currently being evaluated as a cancer therapy in those individuals who are deficient for BRCA1 and BRCA2, which function in the same repair pathway as CHK2.

Drugs targeting CHK2: inhibition or activation?

What about functional CHK2 as a therapeutic target? There is currently a debate over the most appropriate way to target CHK2 in tumours that harbour the wild-type form of this kinase. Among the issues are questions of compound selectivity and the potential for combination with DNA-damaging chemotherapy and radiotherapy. At the cellular level it is necessary to take account of the different molecular lesions caused by different DNA-damaging agents, which elicit distinct responses from the CHK2 pathway. As mentioned previously, the context of intervention is also important, particularly the tumour genetic background such as p53 mutational status. The level of intrinsic DNA damage in a given tumour cell type and the degree to which CHK2 functions might be essential for maintenance of the transformed phenotype have to be considered. In circumstances of high intrinsic DNA damage, a CHK2 inhibitor might have the potential for single-agent efficacy. However, in tumours where activated CHK2 contributes directly to the malignant phenotype or to resistance to DNA-damaging agents, a combination of a CHK2 inhibitor with a DNA-damaging agent might lead to an increased therapeutic index. There are also suggestions that hyperactivation of CHK2 pathways, in the absence of extrinsic DNA-damaging agents, would be effective for inducing certain tumour cells into senescence or even apoptosis.

Inhibition of CHK2. There are several lines of evidence that suggest that CHK2 inhibition in combination with genotoxic agents (IR and chemotherapeutics) might have therapeutic value. Inhibition of CHK2 expression has been found to attenuate DNA damage-induced cell-cycle checkpoints and to enhance apoptotic activity in HEK293 cells[38]. CHK2 inhibition has also shown activity in two cellular models of mitotic catastrophe, indicating a potential for chemosensitization with doxorubicin[46]. In a recent report, targeting of CHK2 using small interfering RNA or a dominant-negative form of this kinase (CHK2-DN) prevented survivin release from the mitochondria and enhanced apoptosis following DNA-damage by IR or doxorubicin[47]. In addition, the conditional expression of CHK2-DN showed potentiation of doxorubicin cytotoxicity in the HCT116 colon carcinoma cell line grown as a xenograft in mice. It should not be overlooked, however, that several studies have reported no therapeutic value of CHK2 inhibition. In particular, it has been reported that loss of CHK2 function has no additional therapeutic benefit compared with loss of CHK1 alone when combined with the antimetabolites 5-fluoro-2′-deoxyuridine and gemcitabine[110]. Similarly, small interfering RNA studies in two cancer cell lines have led to the proposal that CHK1 is the only checkpoint kinase that is relevant as a drug target[111]. In part, this underscores the difficulty of reconciling results from multiple cell line models, using non-equivalent DNA-damaging agents and different strategies for inhibition of CHK2 function. It may be that complete validation of this target in the context of cell killing *in vitro* and *in vivo* must await pharmacological studies with selective small-molecule inhibitors of CHK2 in tumours of defined genetic background.

To date, all reported inhibitors of CHK2 target the ATP-binding pocket of this kinase (TABLE 2). This is not surprising as most compounds have been identified

Box 3 | Is CHK2 required for the p53-mediated DNA-damage response signal?

A number of loss-of-function model systems have been used to assess the requirement of CHK2 for p53 activation and cell-cycle arrest in response to DNA damage. Hirao *et al.* first showed that *Chek2*$^{-/-}$ murine embryonic stem cells are partially defective in late, but not early, G2 arrest following ionizing radiation (IR) (embryonic stem cells do not arrest in G1)[127]. *Chek2*$^{-/-}$ thymocytes, which were generated by injecting *Chek2*$^{-/-}$ chimeras into blastocysts from *Rag1*$^{-/-}$ mice, failed to stabilize p53 and induce p21 and BAX. Mouse embryonic fibroblasts (MEFs) obtained from the same genetic mutants retained their ability to induce G1 arrest and expression of p21 (REF. 128). Using another gene-targeted *Chek2*$^{-/-}$ mouse, Hirao *et al.* showed that CHK2 is not required for S-phase arrest following IR treatment, but is necessary for IR-induced apoptosis and G1 arrest following low, but not high, IR dosage[129]. Analysing a third mouse knockout mutant for CHK2, Takai *et al.* showed that although CHK2 is dispensable for IR-induced S-phase and G2 arrest, it is required for p53-dependent maintenance of G1 arrest, but not for its p53-independent initiation[23]. These *Chek2*$^{-/-}$ cells showed impaired p53 stabilization and transcriptional activity, but demonstrate that CHK2 is not required for IR-induced phosphorylation and acetylation of p53 (REF. 23). This correlates with the ablation of *CHEK2* in the human colon cancer cell line HCT116, which does not inhibit IR-induced phosphorylation and stabilization of p53 and importantly does not prevent cell-cycle arrest[130]. Similarly, small interfering RNA-mediated ablation of *CHEK1* or *CHEK2*, or both, in a number of human cancer cell lines does not prevent accumulation of p53 following radiomimetic DNA damage[131]. These latter studies therefore question the requirement of an ataxia telangiectasia mutated (ATM)–CHK2–p53 pathway for eliciting the response to DNA damage in human cancer cell lines. However, all these studies agree that genetic loss of murine *Chek2* impairs IR-mediated apoptosis and does not predispose to spontaneous tumour development, thus questioning the role of the CHK2-mediated DNA-damage response in preventing tumorigenesis. It is possible that this is the result of intrinsic differences between species and that tumorigenesis prevention is different in mouse and human. This also emphasizes that loss of CHEK2 function may require synergetic association with loss of function in other genes to promote tumour development.

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Table 1 | **Frequencies of founder mutations in different populations***

	Mutation (%)				
Population	**1100delC‡**	**I157T§**	**IVS + 2G>A‖**	**Del5395¶**	**S428F#**
Dutch	1.3–1.6	NA	NA	NA	NA
Finnish	1.1–1.4	5.5	NA	NA	NA
Polish	0.2–0.25	4.8	0.3	0.4	NA
German	0.15–0.25	0.6	0-0.4	NA	NA
American	0.3-0.4	0.9	NA	NA	NA
Australian	0.14	NA	NA	NA	NA
Swedish	0.6–1.0	NA	NA	NA	NA
Byelorussian	NA	1.3	0.2	NA	NA
Ashkenazi Jews	NA	NA	NA	NA	1.37
Czech Republic	0.3	NA	NA	NA	NA
Italy	0.11	NA	NA	NA	NA
Canada	0.2	NA	NA	NA	NA

*Data taken from Refs 4, 5. ‡1100delC, defective in kinase activity[89]. §157T, fails to phosphorylate CDC25A and to bind p53 and BRCA1 (REF. 21,150,151). ‖IVS + 2G>A, results in abnormal splicing, creating a premature termination codon leading to the disruption of protein expression[83]. ¶Del5395, encompasses a deletion of exons 9 and 10 of the gene resulting in a truncated protein with loss of part of the kinase domain[5,7]. #S428F, defective in kinase activity[82]. NA, not available.

through screening for inhibitors of the catalytic activity of this protein. As a result, CHK2 inhibitors tend to inhibit other kinases in addition to their primary target, and in particular the functional relative, CHK1. This has meant that the difficulty of examining the CHK2-dependent pathways separately from CHK1-mediated responses has been a recurrent theme in the pharmacological validation of CHK2 as a drug target. For example, the dual CHK1 and CHK2 inhibitors Go6976 and EXEL-9844 (currently in clinical trials) showed effects in cellular assays that are also seen with loss of CHK1 alone and so it is not possible to ascertain from these studies the independent effect of CHK2 inhibition[112,113].

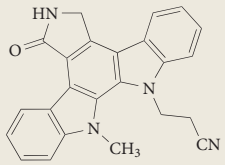
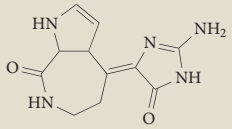
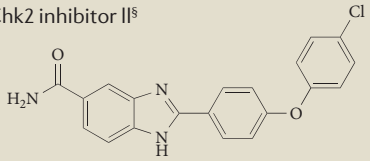
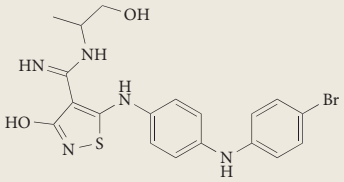
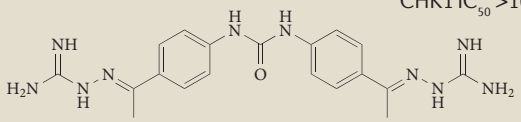
The advent of selective inhibitors of CHK2 promises to address this problem and the recent publication of the X-ray crystal structure of the CHK2 kinase domain will be useful in the development of additional chemical classes of CHK2-selective inhibitors[114]. A recent report identified a series of bis-guanylhydrazone compounds as potent CHK2 inhibitors using *in vitro* kinase assays[115], although these inhibitors did not show inhibition of CHK2 in cells and might suffer from confounding off-target activities or poor cell permeability. In addition, a series of isothiazole carboxamides has provided selective (versus CHK1) CHK2 inhibitors, such as VRX0466617 (REFS 116,117). This small molecule clearly blocked CHK2 activity in cells, but did not significantly change the cell-cycle distribution or prevent the G_2/M arrest in short-term culture of normal or irradiated cells. In longer-term culture (>6 days) exposure of normal cells to VRX0466617 alone led to an antiproliferative effect, but the possibility that this was a manifestation of an off-target activity could not be ruled out. Evaluation of this compound in combination with doxorubicin or cisplatin showed no potentiation of cytotoxicity. It should be noted, however, that the combination studies were performed in the MCF-7 cell line, which harbours wild-type p53. It will therefore be interesting to test whether selective and potent inhibitors of CHK2 potentiate cytotoxic treatments in p53-deficient cell lines.

One therapeutic strategy in which inhibition of CHK2 has clear value is radioprotection. As discussed earlier, targeted disruption of *Chk2* in irradiated mice leads to increased survival through the suppression of apoptosis[23,33–35]. This led to the hypothesis that targeting CHK2 with a small-molecule inhibitor might suppress the side effects of radiotherapy, which is administered to approximately 50% of all cancer patients. The first pharmacological validation of this therapeutic strategy came with a report on a series of selective ATP-competitive 2-arylbenzimidazole CHK2 kinase inhibitors, which were developed as radioprotective agents for normal proliferating tissues[118]. A potent 2-arylbenzimidazole prevented γ-irradiation-induced apoptosis of $CD4^+$ and $CD8^+$ human T cells isolated from blood at doses consistent with the biochemical measurement of CHK2 inhibition. This radioprotective effect has also been reported with the structurally distinct VRX0466617 CHK2 inhibitor in isolated mouse thymocytes[117]. The observation of radioprotection of normal cells with two distinct and selective CHK2 inhibitors suggests that this is a promising therapeutic context to pursue.

Activation of CHK2. As an alternative to CHK2 inhibition, there may also be circumstances where activation of this kinase could have therapeutic value. CHK2 clearly has a role in the barrier to oncogenesis and its activation in the absence of DNA-damaging agents may force tumour cells to exit the proliferative state, either through death or senescence. To date there is only one report investigating this experimental therapeutic approach[50]. This study involved stable transfection of p53-deficient DLD1 colon cancer and HeLa cervical carcinoma cell lines with a tetracycline-inducible form

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Table 2 | **Structures and biological activities of selected CHK2 inhibitors**

Inhibitor	CHK2 potency and selectivity versus CHK1	Cell lines investigated	Inhibitor effects	Refs
Go6976 	Equipotent at CHK2 and CHK1*	MDA-MB-231 breast cancer (p53 mutant) MCF-10A breast cancer (p53 wild type)	Abrogation of the G_2/S phase arrest induced by the topoisomerase I inhibitor SN38 No effect on the SN38-induced G_2/S phase arrest in p53 wild-type cells	112
Debromohymenialdisine 	CHK2 IC_{50} = 183 nM‡ CHK1 IC_{50} = 725 nM	14.3.3δ-deficient HCT116 colon cancer Syncytia arrested in G_2 from fusion of asynchronous HeLa cervical cancer cells	Chemosensitized cells to doxorubicin Provoked mitotic catastrophe	46
EXEL-9844 (structure not disclosed)	CHK2 K_i = 0.07 nM CHK1 K_i = 2.2 nM	PANC-1 pancreatic cancer AsPC1 pancreatic cancer HeLa cervical cancer SKOV-3 ovarian cancer	Abrogation of the G_1 arrest induced by the antimetabolite gemcitabine Chemosensitization of PANC-1 xenografts to gemcitabine	113
Chk2 inhibitor II§ 	CHK2 K_i = 37 nM CHK1 IC_{50} >10,000 nM	$CD4^+$ and $CD8^+$ T cells from human blood	T cells rescued from γ-irradiation-induced apoptosis (inhibitor EC_{50} 3-7 μM)	118
VRX0466617‖ 	CHK2 K_i = 11 nM CHK1 IC_{50} >10,000 nM	HCT116 colon cancer (p53 wild type) Bj-hTERT fibroblasts EBV-immortalized lymphoblastoid LCL-N cells Isolated mouse thymocytes	Prevented CHK2-dependent, γ-radiation-induced degradation of MDMX protein Protected thymocytes from γ-radiation induced apoptosis	117
NSC 109555¶ 	CHK2 IC_{50} = 240 nM CHK1 IC_{50} >10,000 nM	MCF-7 breast cancer (p53 wild type) HT29 (p53 mutant)	No cellular effects when used in combination with topotecan or camptothecin — attributed to confounding off-target activities and poor cell permeability	115

*Inhibition determined in a cell-based assay; enzyme inhibition parameters not described. ‡Inhibition data from REF. 149. §Systematic name: 2-(4-(4-chlorophenoxy)phenyl)-1*H*-benzo[d]imidazole-5-carboxamide. ‖Systematic name: 5-(4-(4-bromophenylamino)phenylamino)-3-hydroxy-*N*-(1-hydroxypropan-2-yl)isothiazole-4-carboximidamide. ¶Systematic name: (2E,2′E)-2,2′-(1,1′-(4,4′-carbonylbis(azanediyl)bis(4,1-phenylene))bis(ethan-1-yl-1-ylidene))bis(hydrazinecarboximidamide). EC_{50}, 50% effective concentration; IC_{50}, 50% inhibitory concentration.

of CHK2 (REF. 50). On induction, CHK2 was produced in both cell lines and activated owing to phosphorylation at T68 in the absence of DNA-damaging agents, perhaps due to a background level of constitutive DNA damage or through oligomerization of the protein[114]. The increased levels of activated CHK2 led to decreased cell proliferation, G_2 arrest and increased apoptosis. Additionally, by eye the cells looked senescent. Thus, manipulation of CHK2 activation, for example, through inhibition of the PPM1D/WIP1 phosphatase (which antagonizes CHK2 activation through T68 dephosphorylation) might have therapeutic benefit. However, a note of caution is needed as this study involved overexpression of CHK2 and so it may be that only the introduction of large quantities of CHK2 into cells and not chemical activation of the endogenous kinase will allow senescence or death.

Altogether, cellular and pharmacological studies demonstrate that the response of a tumour to CHK2 manipulation will depend on a specific cellular context. Given that many tumour cells exhibit high levels of intrinsic DNA damage, the functional availability of the DNA-damage response and repair components will dictate the therapeutic outcomes of CHK2 inhibition or activation. This is exemplified by a recent study showing that pharmacological inhibition of the repair machinery in combination with IR results in increased activation of CHK1 and CHK2 (REF. 119). Consequently, the ability of checkpoint inhibitors to abrogate the G_2 arrest and sensitize these cells to IR was reduced compared with their activity in irradiated cells not exposed to inhibitors of DNA repair. These results highlight the importance of considering the balance of all components in the CHK2 pathway: the

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levels of DNA damage, the degree of signalling through CHK2 and the efficiency of DNA repair in response to checkpoint activation.

Conclusion

The last few years have seen a confluence of the genetics and biology of CHK2. Recent progress highlighted here demonstrates that CHK2 is a versatile and multifunctional kinase that regulates the cell's response to DNA damage by phosphorylating a number of distinct cellular substrates. As such, it might prevent tumour progression by averting genomic instability through DNA repair and, if this is not possible, by causing the cell to senesce or die (FIG. 1). This correlates with human genetic studies clearly showing that *CHEK2* is a multiorgan tumour susceptibility gene[68].

For many inherited cancer-predisposing genes, predictive testing to facilitate early detection is offered to patients when a pathogenic mutation in affected family members is known. Testing individuals for mutations in *CHEK2* may not be appropriate yet as evidence indicates that *CHEK2* alone does not predispose to cancer[61–63,83,88]. In addition, a future challenge will be the molecular profiling of tumours, which could reveal specific genetic characteristics involved in susceptibility to cancer that are associated with *CHEK2* mutations. A potential therapeutic approach in those patients whose tumours harbour *CHEK2* mutations is treatment with inhibitors of other proteins involved in DNA-repair pathways that may be lethal in combination with loss of *CHEK2* (REF. 109). However, this strategy awaits validation.

Looking at the other side of the coin, for those cancer patients with a functional CHK2 protein a key issue for the future is defining how this kinase manages to elicit such distinct cellular outcomes as cell survival through DNA repair versus apoptosis or senescence. This is likely to depend on the genetic background of the cell, in particular its p53 status. Therefore, therapeutic intervention at the level of CHK2 will be influenced by a number of factors, which need to be better defined and taken into consideration. Importantly, the partial overlap and redundancy with CHK1 needs to be addressed further. As mouse models have not yet been able to resolve this issue, the advent of potent and selective CHK2 inhibitors, as discussed in this Review, will provide an additional tool with which to explore it better. Together, this will help define the best therapeutic strategy for a particular tumour, be it exploitation of CHK2 loss, CHK2 inhibition or even activation of this kinase. One thing is for sure: there is still a lot to discover about CHK2, and the next decade of research on this molecule promises to be at least as exciting as the last.

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Competing interests statement
The authors declare competing financial interests: see web version for details.

DATABASES
Entrez Gene: http://www.ncbi.nlm.nih.gov/entrez/query.fcgi?db=gene
AATF | ATM | ATR | BIRC5 | BRCA1 | BRCA2 | CDC25A | CDC25C | *CDKN1B* | CHK1 | CHK2 | MDM2 | MDM4 | p53 | PLK1 | PLK3 | PML | PRKDC

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